



Reliable and Safe Systems

Presented by:

1. Kumar Ankit (1553165)
2. Asmita Nagajibhai Bhuva (1541650)
3. Feneel Niteshkumar Ghevariya (1541874)
4. Kenil Shantilal Dhorajiya (1548729)
5. Manvi Samyak Savani (1542545)
6. Nidhiben Himmatbhai Mandaviya (1548745)
7. Nikunjkumar Damjibhai Sanghani (1547447)

A close-up photograph of mechanical gears, likely from a watch movement, with a blue tint. The gears are intricate, with visible teeth and a central hub. The image is positioned on the left side of the slide, partially overlapping the title area.

Introduction to Reliable and Safe Systems

Reliable and safe systems are the backbone of modern society, ensuring the seamless functioning of critical infrastructure, technology, and devices. These systems are designed to perform consistently and minimize the risk of failure or harm, allowing us to confidently depend on them in our daily lives.

Definition of Reliable Systems

Reliable systems are those that consistently deliver the expected performance, functionality, and service over time. They are designed to withstand various stresses, environmental conditions, and usage patterns without experiencing significant degradation or failure. Reliability is essential for critical applications where downtime or malfunctions can have severe consequences.





Definition of Safe Systems

Safe systems are those designed to minimize the risk of of harm to people, property, or the environment. They They incorporate robust safety features, fail-safe mechanisms, and emergency response protocols to ensure ensure that even in the event of a failure or malfunction, the malfunction, the potential for damage or injury is greatly greatly reduced. Safety is a paramount consideration in consideration in industries such as transportation, healthcare, and energy production.

Differences between Reliable and Safe Systems Systems

Reliability

Focused on consistent performance and minimizing downtime or disruptions.

Safety

Prioritizes the prevention of harm to people, property, and the environment.

Overlap

Many systems require both reliability and safety to function effectively and responsibly.

Common Features of Reliable and Safe Systems

1

Redundancy

Multiple backup components or systems to ensure continuity of operation.

2

Rigorous Testing

Extensive quality assurance and stress testing to identify and mitigate potential failures.

3

Predictive Maintenance

Proactive monitoring and servicing to prevent issues before they arise.

4

Fail-Safe Mechanisms

Automatic fallback systems to ensure a safe state in the event of a failure.

Examples of Reliable Systems

1

Aircraft Navigation

Highly accurate and redundant systems to guide aircraft safely through the skies.

2

Power Grids

Interconnected networks of power generation, transmission, and distribution to ensure continuous energy supply.

3

Water Treatment Plants

Robust systems that reliably purify and distribute clean water to communities.



Examples of Safe Systems



Automated Safety Systems

Sensors, controls, and emergency procedures to protect people in hazardous environments.



Cybersecurity Measures

Robust encryption, access controls, and threat detection to safeguard digital systems.



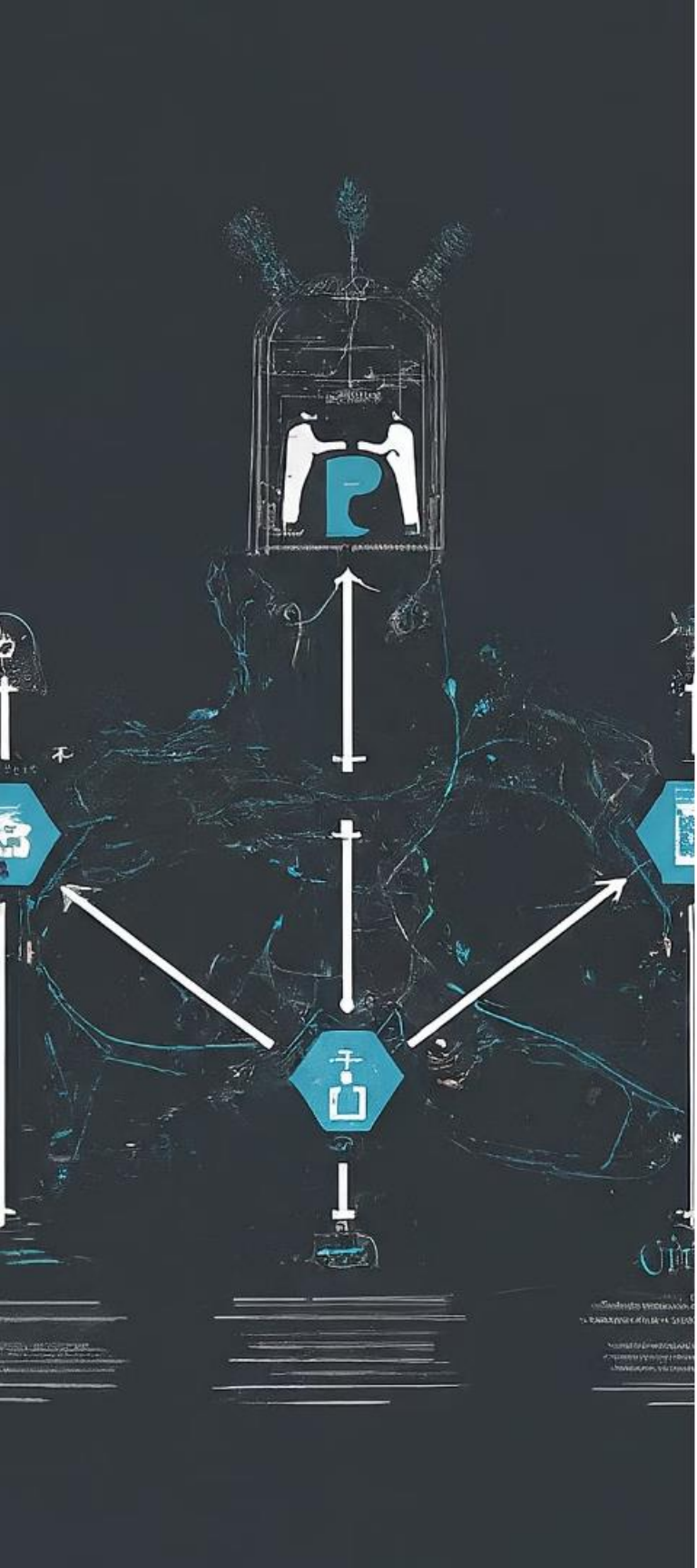
Emergency Response Plans

Detailed protocols and training to ensure a coordinated and effective response to crises.



Hazard Mitigation Strategies

Proactive measures to identify and minimize potential risks in high-stakes environments.



Conclusion and Key Takeaways

1

Reliable and Safe

Reliable and safe systems are essential for modern society, ensuring consistent performance and minimizing risks.

2

Common Features

Reliability and safety often share key features, such as redundancy, testing, maintenance, and fail-safe mechanisms.

3

Critical Applications

Reliable and safe systems are crucial in industries like transportation, healthcare, energy, and infrastructure.

Thank You 😊